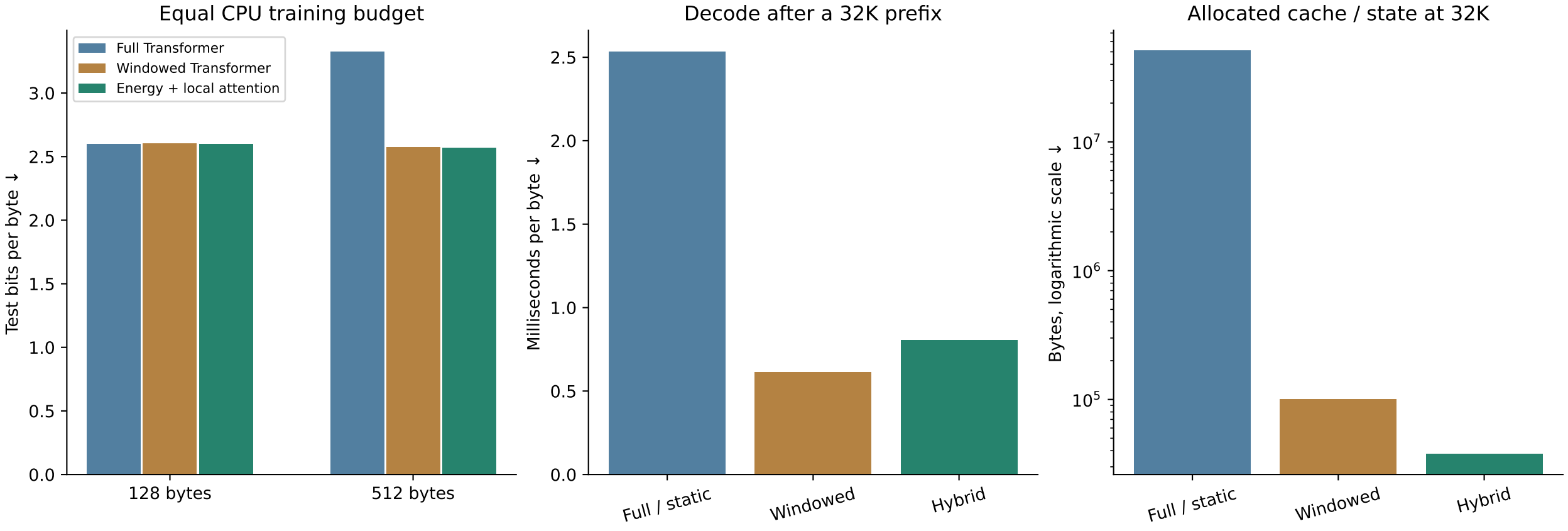
Equal CPU training budget
Full Transformer
Windowed Transformer
Energy + local attention
Test bits per byte ↓
3.0
2.5
2.0
1.5
1.0
0.5
0.0
128 bytes
512 bytes
Decode after a 32K prefix
Milliseconds per byte ↓
2.5
2.0
1.5
1.0
0.5
0.0
Full / static
Windowed
Hybrid
Allocated cache / state at 32K
Bytes, logarithmic scale ↓
10^7
10^6
10^5
Full / static
Windowed
Hybrid